Daffodil International University

DEPARTMENT OF INFORMATION & COMMUNICATION ENGINEERING

Dhaka, BANGLADESH.

COURSE NO.: ICE 3242

COURSE TITLE: DIGITAL SIGNAL PROCESSING Lab

Experiment No. 05

Name of Experiment: Discrete-Time Convolution and Correlation

Objectives:

- To find the convolution of two sequences using `conv()` and using a simple for-loop.
- To find the length and starting index of the convolution output.
- To compute autocorrelation and cross-correlation using `xcorr()`.
- To estimate the delay of a noisy signal using cross-correlation.

THEORY:

1. Convolution

The output of a linear time-invariant (LTI) system is the **convolution** of the input x(n) with the system's impulse response h(n):

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

For each output sample y(n), four steps are repeated:

1. **Fold** h(k) to get h(−k).
2. **Shift** it by n to get h(n − k).
3. **Multiply** x(k) and h(n − k) sample by sample.
4. **Sum** all the products to get y(n).

In this experiment x(n) = {1, 2, 3, 1} for $-1 \le n \le 2$ and h(n) = {3, 2, 1} for $0 \le n \le 2$. Fig. 5.1 shows the four steps for n = 1, giving y(1) = 1·1 + 2·2 + 3·3 = 14.

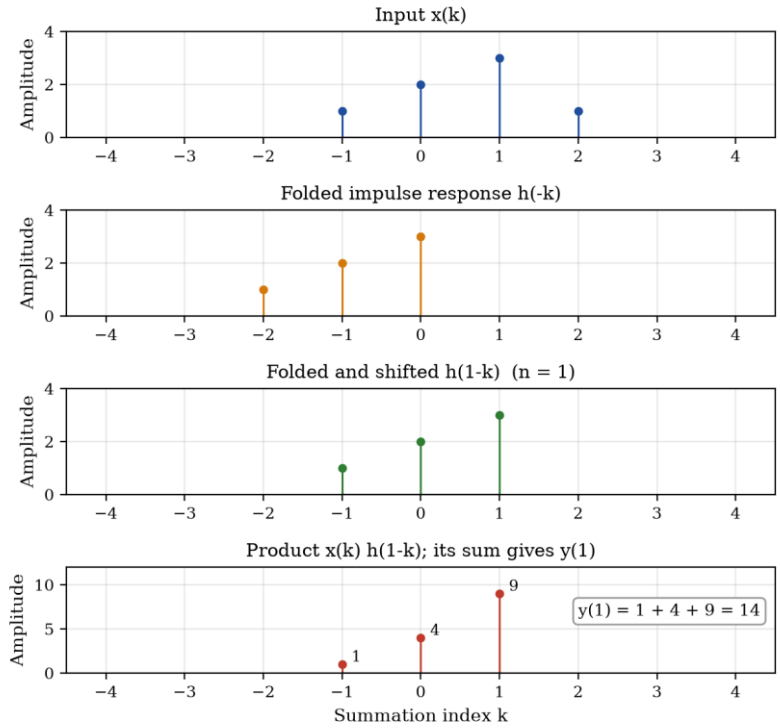


Fig. 5.1 *Fold, shift, multiply and sum for n = 1*

Table 5.1 Step-by-step convolution of x(n) and h(n)

n	Calculation	y(n)
−1	1·3	3
0	1·2 + 2·3	8
1	1·1 + 2·2 + 3·3	14
2	2·1 + 3·2 + 1·3	11
3	3·1 + 1·2	5
4	1·1	1

Length and start of the output. If x(n) has N_1 samples starting at n_x0 and h(n) has N_2 samples starting at n_h0, then:

$$N_y = N_1 + N_2 - 1, \qquad n_{y0} = n_{x0} + n_{h0}, \qquad n_{y1} = n_{x1} + n_{h1}$$

Here N_y = 4 + 3 − 1 = 6 and the output starts at n = −1 + 0 = −1, so y(n) = {3, 8, 14, 11, 5, 1} for −1 ≤ n ≤ 4. Note: `conv()` returns only the values; the time index must be built separately.

An easy way to think about it: each input sample x(k) adds one copy of h(n), multiplied by x(k) and shifted to position k. Adding all these copies gives y(n). Program 2 uses exactly this idea (see Table 5.3).

2. Properties of Convolution

Table 5.2 Main properties

Property	Statement	Meaning
Commutative	x * h = h * x	Order does not matter
Associative	(x * h_1) * h_2 = x * (h_1 * h_2)	Two systems in cascade = one system

Property	Statement	Meaning
Distributive	$x * (h_1 + h_2) = x * h_1 + x * h_2$	Two systems in parallel = one system
Identity	$x(n) * \delta(n) = x(n)$	$\delta(n)$ leaves the signal unchanged

3. Correlation

Correlation measures how **similar** two signals are when one is shifted by a lag l. The **cross-correlation** is:

$$r_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n)\, y(n-l) = \sum_{n=-\infty}^{\infty} x(n+l)\, y(n)$$

When x = y it is called the **autocorrelation**. It is symmetric, and its maximum is at l = 0, where it equals the signal energy:

$$r_{xx}(l) = \sum_{n} x(n)\, x(n-l), \qquad r_{xx}(0) = \sum_{n} x^2(n) = E_x, \qquad r_{xx}(-l) = r_{xx}(l)$$

For x(n) = {1, 2, 3, 1}: r_xx(l) = {1, 5, 11, 15, 11, 5, 1} for $-3 \le l \le 3$, and E_x = 1 + 4 + 9 + 1 = 15.

Correlation is the same as convolution **without folding**. So it can also be computed by convolving with a flipped sequence:

$$r_{xy}(l) = x(l) * y(-l)$$

In MATLAB: `xcorr(x, y)` and `conv(x, fliplr(y))` give the same answer.

4. Application: Delay Estimation

A signal s(n) is sent and a delayed, noisy copy r(n) is received. We slide s(n) along r(n) using cross-correlation. Where they match best, the correlation is **largest**, and that lag is the **delay D**:

$$\hat{D} = \arg\max_{l}\ r_{rs}(l)$$

MATLAB commands used

- `conv(x,h)` – convolution of x and h.
- `xcorr(x,y)` – cross-correlation of x and y. `xcorr(x)` – autocorrelation of x.
- `fliplr(x)` – reverses a vector (folding).
- `zeros(1,N)` – a row of N zeros.
- `[value, pos] = max(v)` – the largest value of v and its position.
- `randn(1,N)` – N random noise samples.
- `stem, subplot, title, xlabel` – plotting commands.

PROCEDURE:

Apparatus Required

- System with MATLAB 7.0 and the Signal Processing Toolbox (for `xcorr`).

Algorithm

1. Clear the command window, workspace and figures.
2. Enter the sequences x(n) and h(n).
3. **Program 1:** find y = conv(x, h) and plot x, h and y.

4. **Program 2:** find the same convolution with a simple for-loop.
5. **Program 3:** find the autocorrelation of x(n) with xcorr().
6. **Program 4:** find the cross-correlation with xcorr() and with conv(x, fliplr(y)).
7. **Program 5:** delay a signal by 10 samples, add noise and find the delay from the correlation peak.
8. Note the outputs and paste the figures in the RESULTS section.

Program

1. Convolution Using conv()

```
clc; clear all; close all;

x = [1 2 3 1];      % input signal
h = [3 2 1];        % impulse response

y = conv(x, h)      % output: 3 8 14 11 5 1

nx = -1:2;          % x starts at n = -1
nh = 0:2;           % h starts at n = 0
ny = -1:4;          % y starts at -1+0 = -1 and ends at 2+2 = 4

subplot(3,1,1); stem(nx, x); title('Input x(n)');
subplot(3,1,2); stem(nh, h); title('Impulse response h(n)');
subplot(3,1,3); stem(ny, y); title('Output y(n)');
```

2. Convolution Using a for-Loop

```
clc; clear all; close all;

x = [1 2 3 1];
h = [3 2 1];
y = zeros(1, 6);        % output has 4 + 3 - 1 = 6 samples

for k = 1:4             % take each input sample one by one
    y(k:k+2) = y(k:k+2) + x(k)*h;   % add a scaled, shifted copy of h
end

y                       % 3 8 14 11 5 1 (same as conv)
stem(y); title('Convolution using for-loop');
```

Table 5.3 shows how y is built, one input sample at a time:

Table 5.3 How the for-loop builds y

k	x(k) × h	Added at positions	y after this step
1	1 × [3 2 1] = [3 2 1]	1 to 3	[3 2 1 0 0 0]
2	2 × [3 2 1] = [6 4 2]	2 to 4	[3 8 5 2 0 0]
3	3 × [3 2 1] = [9 6 3]	3 to 5	[3 8 14 8 3 0]
4	1 × [3 2 1] = [3 2 1]	4 to 6	[3 8 14 11 5 1]

3. Autocorrelation

```
clc; clear all; close all;

x = [1 2 3 1];
r = xcorr(x)          % 1 5 11 15 11 5 1
lag = -3:3;
stem(lag, r); title('Autocorrelation of x(n)'); xlabel('lag');
```

```
E = sum(x.^2)          % energy = 15 = middle (largest) value of r
```

4. Cross-Correlation

```
clc; clear all; close all;

x = [1 2 3 1];
y = [1 1 2 1];

r1 = xcorr(x, y)            % 1 4 8 10 7 4 1
r2 = conv(x, fliplr(y))     % same answer
lag = -3:3;

subplot(2,1,1); stem(lag, r1); title('Using xcorr');
subplot(2,1,2); stem(lag, r2); title('Using conv with flipped y');
```

5. Delay Estimation

```
clc; clear all; close all;

s = [1 2 3 2 1];                    % signal we send
r = [zeros(1,10) s zeros(1,5)];     % received: s delayed by 10 samples
r = r + 0.3*randn(1,20);            % add some noise

[c, lag] = xcorr(r, s);             % compare r and s at every lag
[peak, pos] = max(c);               % find the largest value
delay = lag(pos)                    % answer: 10

subplot(3,1,1); stem(s); title('Sent signal s(n)');
subplot(3,1,2); stem(r); title('Received signal r(n)');
subplot(3,1,3); stem(lag, c); title('Cross-correlation');
```

Tip: a line without a semicolon (;) prints its result in the command window.

Program Explanation

MATLAB statement	What it does
`clc; clear all; close all;`	Clears the screen, variables and figure windows.
`y = conv(x, h)`	Finds the convolution: [3 8 14 11 5 1].
`ny = -1:4;`	Time index of y. Start = −1 + 0, end = 2 + 2.
`y = zeros(1, 6);`	Makes an empty output of 6 zeros.
`y(k:k+2) = y(k:k+2) + x(k)*h;`	Multiplies h by x(k) and adds it at positions k to k+2.
`r = xcorr(x)`	Autocorrelation of x.
`conv(x, fliplr(y))`	Correlation done as convolution with flipped y.
`r = [zeros(1,10) s zeros(1,5)];`	Puts 10 zeros before s, i.e. delays it by 10 samples.
`r = r + 0.3*randn(1,20);`	Adds random noise to the received signal.
`[peak, pos] = max(c);`	Finds the largest correlation value and where it is.
`delay = lag(pos)`	The lag at the peak is the delay.

Precautions

- conv() gives only values — the time index of the output must be made separately.

- The output length is $N_1 + N_2 - 1$.
- conv(x, y) is not correlation; use xcorr(x, y) or conv(x, fliplr(y)).
- Noise is random, so the Program 5 figure will look a little different each time, but the delay must still be 10.

RESULTS:

Run each program in MATLAB and paste the figure in the space below.

S.No.	Program	Output Waveform (paste figure here)
1	Convolution using conv()	
2	Convolution using for-loop	
3	Autocorrelation	
4	Cross-correlation	
5	Delay estimation	

Expected Output

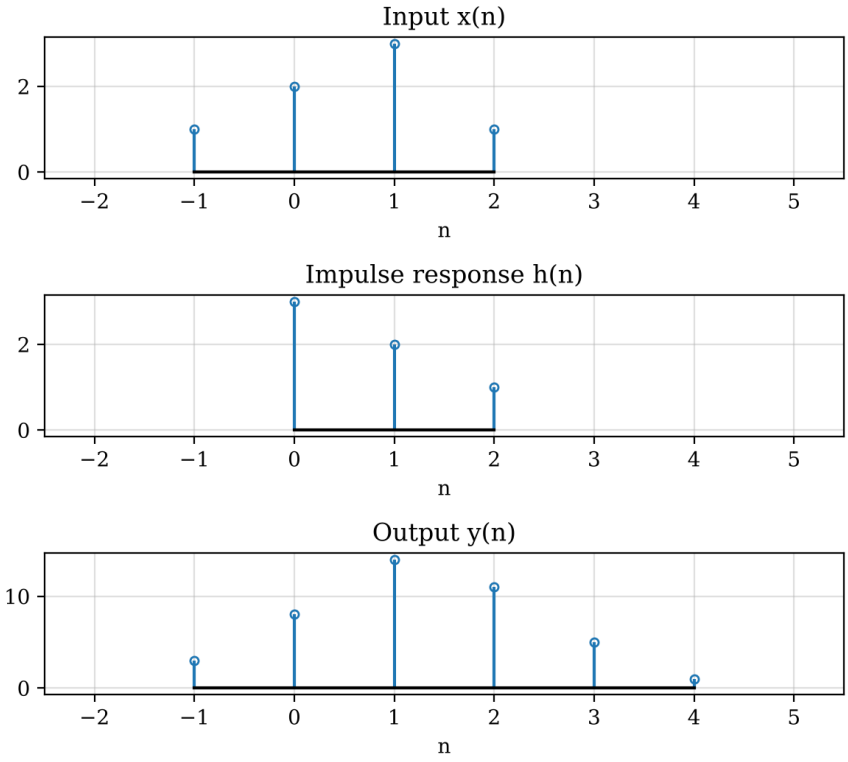


Fig. 5.2 *Program 1 – x(n), h(n) and y(n) = {3, 8, 14, 11, 5, 1} on −1 ≤ n ≤ 4*

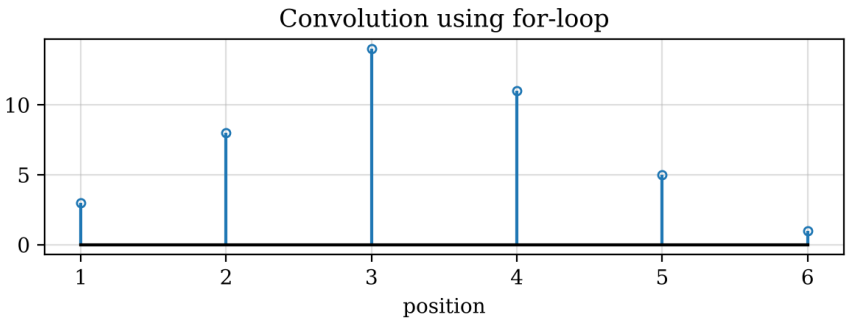


Fig. 5.3 *Program 2 – same values as conv()*

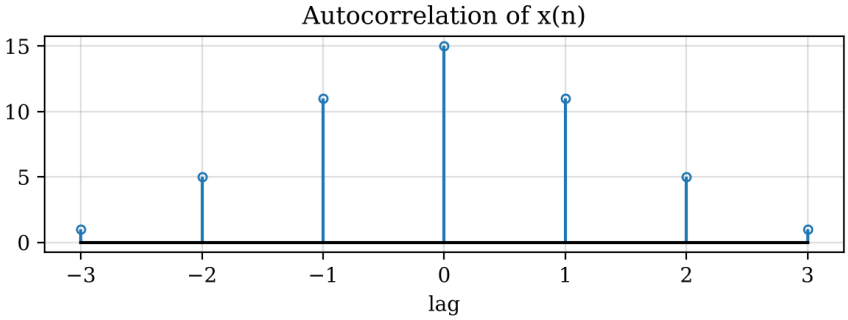


Fig. 5.4 *Program 3 – symmetric, largest value 15 = E_x at lag 0*

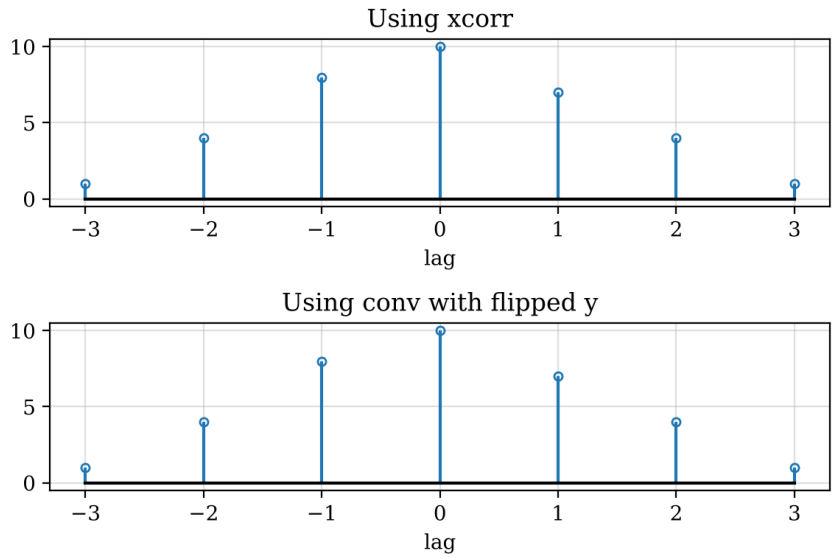


Fig. 5.5 *Program 4 – xcorr and conv(x, fliplr(y)) give the same values*

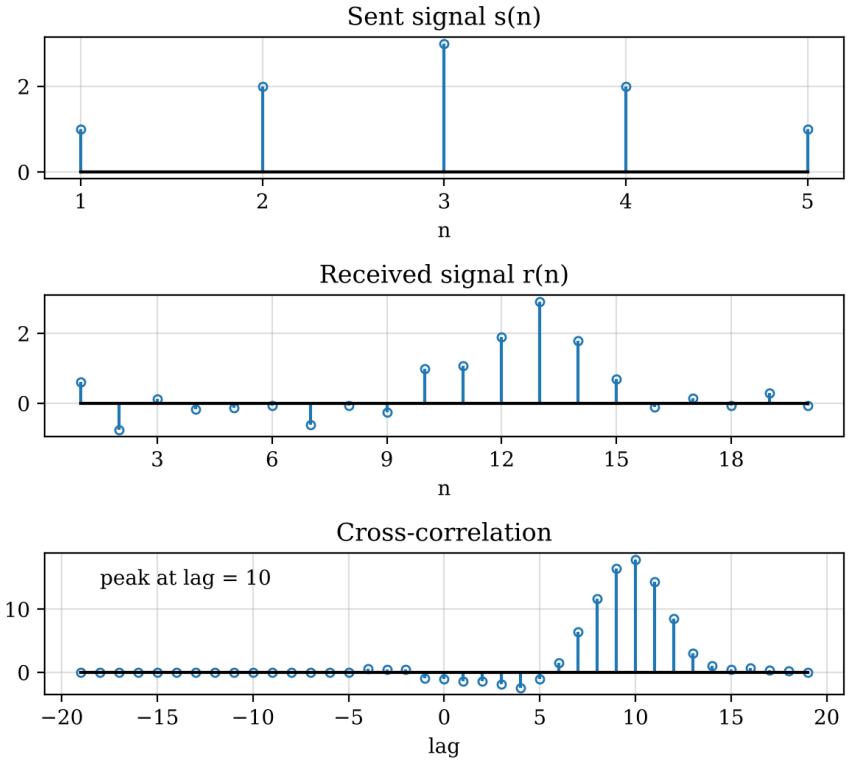


Fig. 5.6 *Program 5 – the correlation peak is at the delay, lag = 10*

Table 5.4 Results

Quantity	**Expected value**	**Observed**
y = conv(x, h)	3, 8, 14, 11, 5, 1	
Index range / length of y	−1 to 4 / 6	
For-loop result	3, 8, 14, 11, 5, 1	
Autocorrelation xcorr(x)	1, 5, 11, 15, 11, 5, 1	
Energy E = sum(x.^2)	15	

Quantity	Expected value	Observed
xcorr(x, y)	1, 4, 8, 10, 7, 4, 1	
conv(x, fliplr(y))	1, 4, 8, 10, 7, 4, 1	
Estimated delay	10	

Observations

Conclusion

The convolution of x(n) = {1, 2, 3, 1} and h(n) = {3, 2, 1} was computed using conv() and a for-loop, both giving y(n) = {3, 8, 14, 11, 5, 1} on $-1 \leq n \leq 4$. The autocorrelation was found to be symmetric with its maximum equal to the signal energy (15), cross-correlation was shown to equal convolution with a flipped sequence, and the delay of a noisy signal was correctly estimated from the cross-correlation peak using MATLAB.

Exercise:

1. Convolve x(n) = {1, 1, 1, 1, 1} with h(n) = {1, 2, 1} using conv(). Plot all three signals.
2. Repeat Exercise 1 using the for-loop program. (Hint: change 6, 4 and k+2 to match the new lengths.)
3. Find the autocorrelation of s = [1 2 3 2 1] using xcorr(s) and using conv(s, fliplr(s)). Are they the same?
4. In Program 5, change zeros(1,10) to zeros(1,5) and randn(1,20) to randn(1,15). Then increase the noise from 0.3 to 1. Is the delay still found correctly?

Viva Questions:

1. What is convolution? Write the convolution sum.
2. What are the four steps of convolution?
3. What is the length of the output of conv(x, h)?
4. Why does conv() not give the time index of the output?
5. What is the difference between convolution and correlation?
6. Why is the autocorrelation maximum at l = 0?
7. How is cross-correlation used to estimate a delay?